

CMOS DIGITAL GATE DESIGN & PERFORMANCE ANALYSIS

Transistor-Level Implementation, Cadence Virtuoso Simulation, Power Analysis and Propagation-Delay Evaluation

Project Focus

CMOS Inverter • 2-Input NAND • 2-Input NOR

Project Objective

To design and evaluate fundamental CMOS logic gates at transistor level using Cadence Virtuoso, verify their Boolean functionality through transient simulations, and quantitatively compare average power and propagation delay to understand practical power–speed trade-offs.

Why this project matters

Digital IC performance is not determined by logic correctness alone. A useful CMOS design must also balance switching speed and power consumption. This project demonstrates that trade-off using actual Cadence simulation measurements rather than only theoretical logic diagrams.

Executive Result

Gate	Average Power	tPHL	tPLH	Mean Displayed Delay
CMOS Inverter	424.1 nW	1.25 μ s	2.05 μ s	1.65 μ s
2-input NAND	227.7 nW	3.27 μ s	0.0272 μ s (27.2 ns)	1.6486 μ s
2-input NOR	283.8 nW	2.848 μ s	2.721 μ s	2.7845 μ s

1. CMOS INVERTER

Structure: one PMOS pull-up transistor and one NMOS pull-down transistor. Their gates share the input and their drains form the output node.

Operating principle: When the input is LOW, the PMOS conducts and the output is HIGH. When the input is HIGH, the NMOS conducts and the output is LOW. Thus, the circuit implements the NOT operation.

Cadence verification: The transient waveform shows the output complementing the input, with approximately 1.8 V at the HIGH level and approximately 0 V at the LOW level. The project also evaluates switching power and marker-based propagation delay.

Measured result: Average power = 424.1 nW; tPHL = 1.25 μ s; tPLH = 2.05 μ s; mean displayed delay = 1.65 μ s.

2. 2-INPUT CMOS NAND GATE

Structure: two PMOS transistors in parallel in the pull-up network and two NMOS transistors in series in the pull-down network.

Operating principle: The output becomes LOW only when both inputs are HIGH. A LOW on either input creates a pull-up path through the PMOS network, keeping the output HIGH.

Cadence verification: The transient A, B and Y waveforms verify NAND functionality. Power and delay marker measurements are used for quantitative performance analysis.

Measured result: Average power = 227.7 nW; tPHL = 3.27 μ s; tPLH = 0.0272 μ s (27.2 ns); mean displayed delay = 1.6486 μ s.

3. 2-INPUT CMOS NOR GATE

Structure: two PMOS transistors in series in the pull-up network and two NMOS transistors in parallel in the pull-down network.

Operating principle: The output is HIGH only when both inputs are LOW. If either input becomes HIGH, the parallel NMOS network provides a pull-down path.

Cadence verification: The transient waveforms confirm NOR functionality, while the power and delay plots provide the performance data.

Measured result: Average power = 283.8 nW; tPHL = 2.848 μ s; tPLH = 2.721 μ s; mean displayed delay = 2.7845 μ s.

4. PERFORMANCE ANALYSIS

Power consumption is compared using Cadence average-power results under the supplied simulation conditions. The measured ranking is:

NAND < NOR < Inverter

The 2-input NAND has the lowest measured average power (227.7 nW), followed by NOR (283.8 nW), while the inverter has the highest measured value (424.1 nW). These values are specific to the supplied transistor models, dimensions, input activity, loading and simulation setup.

5. PROPAGATION DELAY ANALYSIS

Propagation delay is the time difference between corresponding input and output transition reference points. Because rising and falling transitions can differ, tPHL and tPLH are evaluated separately before obtaining the mean displayed delay.

Gate	Calculation	Result
Inverter	tPHL = 10.02213 – 10.02088	1.25 μ s
	tPLH = 15.07907 – 15.07702	2.05 μ s
	Mean = (1.25 + 2.05)/2	1.65 μ s
NAND	tPHL = 14.01825 – 14.01498	3.27 μ s
	tPLH = 3.5550170 – 3.5549898	0.0272 μ s = 27.2 ns
	Mean displayed delay	1.6486 μ s
NOR	tPHL = 12.537702 – 12.534854	2.848 μ s
	tPLH = 5.015139 – 5.012418	2.721 μ s
	Mean = (2.848 + 2.721)/2	2.7845 μ s

6. POWER–DELAY TRADE-OFF

The results show that there is no single gate with the lowest measured power and lowest measured delay simultaneously. The inverter is the speed-oriented result in the supplied report, with the lowest mean displayed delay. The NAND gate is the power-oriented result, with the lowest measured average power. The NOR gate has the highest mean displayed delay in this dataset.

Engineering takeaway: CMOS gate selection should be driven by application requirements. Low-power applications may favor the NAND result, while speed-sensitive designs may favor the inverter result under these particular simulation conditions. A balanced design requires considering both metrics together.

7. PROJECT WORKFLOW

Design → Transistor-Level Schematic → Transient Simulation → Functional Verification → Average-Power Measurement → tPHL/tPLH Measurement → Comparative Analysis → Design Insight

This workflow highlights that the project is not simply a collection of schematics; it is a measurement-driven study connecting transistor-level CMOS implementation with system-level performance considerations.

8. MENTOR-LEVEL PROJECT HIGHLIGHTS

- **Industry-relevant EDA exposure:** Cadence Virtuoso is used for transistor-level schematic and simulation analysis.
- **Design verification:** Boolean functionality is validated through transient waveforms.
- **Quantitative evaluation:** Average power, tPHL and tPLH are extracted from simulation results.
- **Comparative thinking:** Inverter, NAND and NOR are compared using the same performance framework.
- **Engineering insight:** The project demonstrates the practical trade-off between switching speed and power consumption.

9. CONCLUSION

The Cadence Virtuoso simulations demonstrate successful transistor-level implementation and functional verification of a CMOS inverter, 2-input NAND and 2-input NOR gate. The measured results provide a practical comparison of power dissipation and propagation delay. Under the supplied simulation conditions, NAND achieves the lowest measured average power, while the inverter has the lowest mean displayed delay. Overall, the project establishes a clear foundation for understanding CMOS performance optimization and power–speed trade-offs in digital VLSI design.

Note: All numerical values and conclusions in this redesigned report are based on the supplied project document and its reported Cadence measurements.